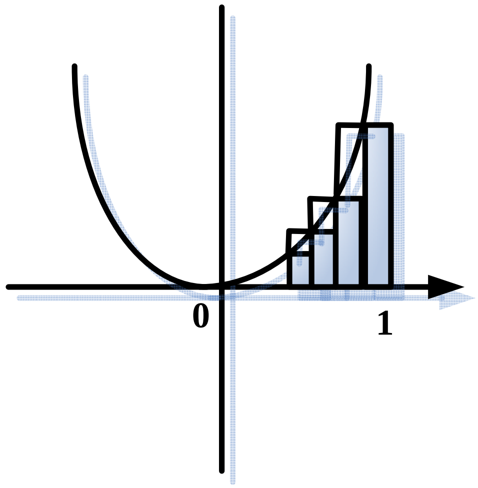
§5-1 Areas and Distances

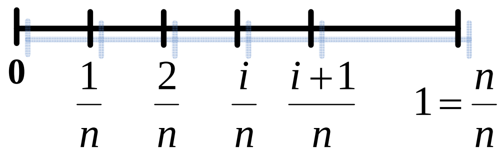
Homework：17,20,21,26.

How to find the area of a region whose boundary does not consist of just line segments ?

Example 1：The area under the curve $f(x)=x^2$ on $[0,1]$.

想法：將此區域分成許多的小長方形，利用夾擊。





把 $[0,1]$ 分成 n 等份。假設 $S=\text{exact area}$

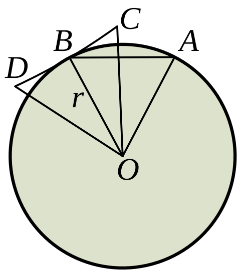
因為 f 在 $[0,1]$ 是 $\nearrow$ ，所以 $\sum_{i=0}^{n-1}\frac{1}{n}f\left(\frac{i}{n}\right)<S<\sum_{i=0}^{n-1}\frac{1}{n}f\left(\frac{i+1}{n}\right)$.

但 $\sum_{i=0}^{n-1}\frac{1}{n}f\left(\frac{i}{n}\right)=\sum_{i=0}^{n-1}\frac{i^2}{n^3}=\frac{(n-1)(n)(2n-1)}{6n^3}\to\frac{1}{3}\ as\ n\to\infty.$

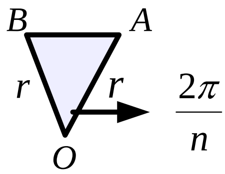
$$\sum_{i=0}^{n-1}\frac{1}{n}f\left(\frac{i+1}{n}\right)=\sum_{i=0}^{n-1}\frac{(i+1)^2}{n^3}=\sum_{i=1}^{n}\frac{i^2}{n^3}=\frac{n(n+1)(2n+1)}{6n^3}\to\frac{1}{3}\ as\ n\to\infty.$$

$$\Rightarrow S=\frac{1}{3}$$

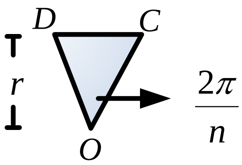
Example 2：



想法：將圓形分為許多小三角形，且利用夾擊。



下界：內接正 n 多邊形(S_n)



上界：外切正 n 多邊形(T_n)

$$S_n = \frac{n}{2} r^2 \sin\frac{2\pi}{n} \to \pi r^2 \quad as\ n \to \infty$$

$$T_n = nr^2 \tan\frac{\pi}{n} \to \pi r^2 \quad as\ n \to \infty$$

$$\Rightarrow A = \pi r^2$$

Summary：irregular area $= \sum$ regular areas

$= \sum$ 小長方形

$= \sum$ 小三角形.

Example 3：

Determine a region whose area is equal to the given limit as follows：

$$\lim_{n\to\infty}\sum_{i=1}^{n}\frac{2}{n}\left(5+\frac{2i}{n}\right)^{10}.$$

Solution：